

Rung One: A Friendly Guide

Companion notes to “A Machine-Verified Exclusion of an Automatic Collatz Itinerary by Mahler’s M

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1 Where we were

The Collatz game: take a whole number; if it is even, halve it; if it is odd, triple it, add one, and halve. The conjecture says you always reach 1. Nobody can prove it.

Earlier papers in this series set up a *diary* for each orbit: write 1 every time the number is odd and 0 every time it is even. Two things are known about diaries. First, the diary determines the starting number completely — no two starting numbers write the same diary. Second, if an orbit is going to run off to infinity (the thing the conjecture forbids), its diary has to contain a lot of ones: more than about 63% of the entries, forever. Fewer ones means more halvings, and the number shrinks.

So one way to attack the conjecture is to go through the possible diaries, from simplest to most complicated, and rule them out one class at a time. This is the *ladder*. The bottom rung was already proved: a diary that eventually repeats itself belongs to an orbit that eventually repeats itself, which is bounded, so it cannot be a runaway.

The next rung up is the simplest kind of diary that never repeats. Take the rule

$$1 \rightarrow 110, \quad 0 \rightarrow 011,$$

start from a single 1, and apply the rule over and over:

$$1 \rightarrow 110 \rightarrow 110110011 \rightarrow 110110011110110011011 \rightarrow \dots$$

The limit is an infinite word w that never repeats but is completely structured: every third letter is a 1, and every block of three contains exactly two ones. Two-thirds ones is more than 63%, so nothing we knew forbade a runaway orbit from keeping exactly this diary. The earlier paper boiled the question down to one number.

The one number. Think of w as the coefficients of a power series $F(z) = w_0 + w_1z + w_2z^2 + \dots$. There is a unique “number” whose diary is w — but it is a number in the 2-*adic* sense, a kind of infinite binary expansion running off to the left. It equals $-10 + \frac{5}{9}F_2(8/9)$, where $F_2(8/9)$ means: plug $z = 8/9$ into the series and add it up 2-adically (where $8/9$ is small because 8 is divisible by a high power of 2). An ordinary

*Written in collaboration with Claude (Anthropic). These notes explain the ideas; the paper and the Lean file contain the proofs. The Collatz conjecture remains open.

whole number has a binary expansion that stops. So the question “does any whole number have diary w ?” is the same as the question “is $F_2(8/9)$ a fraction?”

The earlier paper also showed the obvious attack fails. If you cut the series off after N terms you get a fraction with denominator 9^N that agrees with $F_2(8/9)$ to about $3N$ binary digits. To prove a number is not a fraction by approximation, the agreement has to grow faster than the denominator, and 2^{3N} versus 9^N is a losing race — the ratio is 0.946, just under 1. (And 0.946 is, once again, the 63% threshold in disguise.)

2 What is new

We prove $F_2(8/9)$ is not a fraction. Therefore no whole number has diary w , and none has a diary that is w with the first few entries torn off either. No Collatz orbit ever falls into this pattern. This is the first time a non-repeating diary has been ruled out, and every step of the proof has been checked by the Lean proof assistant.

The trick that does it is old — it goes back to Kurt Mahler in 1929 — and the honest summary is that the earlier paper overestimated how much of Mahler’s machinery would be needed. Here is the idea.

3 The idea in three moves

Move 1: a better approximation

Instead of cutting the series off, find two small polynomials p_0 and p_1 so that $p_0(z) + p_1(z)F(z)$ starts with a very high power of z . This is a linear-algebra problem: $p_0 = 1 + z^3 - z^4$ and $p_1 = -1 + z - z^2 + z^3$ have ten coefficients between them, and we can ask for nine conditions — the coefficients of z^0 through z^8 in $p_0 + p_1F$ all vanish. They do, and the coefficient of z^9 turns out to be -1 :

$$p_0(z) + p_1(z)F(z) = -z^9 + z^{10} + z^{12} + \dots$$

This depends only on the first ten letters of w , and a computer checks it in a blink.

Now plug in $z = 8/9$ 2-adically. Because $8/9$ is 2-adically tiny and all the coefficients are whole numbers, the right-hand side is *exactly* as small as its first term: it is divisible by 2^{27} (that is 8^9) and not by 2^{28} . There is no estimating and no error term; it is exact. This is the point where the 2-adic world is kinder than the ordinary one. In Mahler’s original setting one has to work hard to bound this quantity and then work harder to show it is not accidentally zero. Here both come for free.

Move 2: the functional equation, and the tower

The rule $1 \rightarrow 110, 0 \rightarrow 011$ that builds w translates into an equation for F :

$$F(z) = \frac{z + z^2}{1 - z^3} + (1 - z^2) F(z^3).$$

It says that F at z is determined by F at z^3 , and vice versa. So if $F_2(8/9)$ were a fraction, then $F_2((8/9)^3)$ would be a fraction, and $F_2((8/9)^9)$, and so on up a tower of points $(8/9)^{3^k}$

that get 2-adically smaller and smaller — and at each level the fraction gets more complicated. We can keep exact track of how complicated: at level k the numerator and denominator are at most about $9^{(5/2) \cdot 3^k}$.

Move 3: the collision

Now run Move 1 at level k instead of level 0. The Padé expression $p_0 + p_1 F$ evaluated at $(8/9)^{3^k}$ is divisible by $2^{27 \cdot 3^k}$ exactly. But if F_2 at that point were a fraction u/D , the same expression would be a fraction whose numerator we can bound: the polynomials contribute at most $9^{4 \cdot 3^k}$ and the tower contributes at most $9^{3 \cdot 3^k}$ or so, for a total of roughly $9^{7 \cdot 3^k}$ times a constant. A nonzero whole number divisible by $2^{27 \cdot 3^k}$ is at least $2^{27 \cdot 3^k}$. So we need

$$2^{27 \cdot 3^k} \leq (\text{constant}) \cdot 9^{7 \cdot 3^k},$$

and since $2^{27} = 134,217,728$ is bigger than $9^7 = 4,782,969$ by a factor of 28, the left side wins by a factor of 28^{3^k} , which beats any constant once k is large. Contradiction. So $F_2(8/9)$ is not a fraction.

Why this beats the 0.946 barrier. Cutting off the series costs a factor of 9 in the denominator for every factor of 8 in 2-adic smallness: ratio 0.946. The Padé trick costs a factor of 9^4 (the degree of the polynomials) for a factor of 8^9 (the order of vanishing): ratio 1.89. Using the functional equation to move up the tower multiplies both sides by 3 at each step, so the ratio is preserved while a fixed overhead is left behind. That squaring of the approximation quality is exactly what Mahler’s method is for; the earlier paper knew this, but assumed the full transcendence machinery would be needed. Only the smallest piece of it is.

4 What was checked, and how

The Lean file `Collatz/Rung1.lean` contains no 2-adic numbers at all. Every statement above about “2-adically small” became a statement “divisible by $2^{\text{something}}$ ” about ordinary integers, and every statement about the series became a statement about its first L terms, with L arbitrary. The chain is:

1. The word w is defined by its rule, and its first ten letters are computed.
2. The functional equation is proved as an exact identity between finite sums (no limits).
3. A whole number with diary w — or w with s letters torn off — forces a divisibility condition for every prefix length (this uses the exact orbit identity from the earlier papers).
4. The divisibility condition climbs the tower.
5. The Padé certificate is checked, and its exact 2-adic size is extracted as a divisibility-plus-oddness statement.
6. The sizes are bounded, and $2^{27} > 28 \cdot 9^7$ does the rest.

Lean’s kernel accepts the proof using only its three standard foundational axioms. There is no numerical computation to trust, no external tool, and no **sorry**.

5 What this does and does not say

It does not say anything about the Collatz conjecture in general. It rules out one family of diaries — one infinite word and all its tails — out of uncountably many. The value of the result is that it is the first rung where the diary was genuinely non-repeating, and that the proof pattern is a template: any word built by a fixed-length substitution rule in which every block has the same number of ones gives the same shape of functional equation and the same shape of certificate, so the same proof should run.

It also does not claim a new theorem in number theory. That p -adic values of Mahler functions at small rational points are irrational is a known kind of fact; experts would not be surprised by $F_2(8/9) \notin \mathbb{Q}$. What is new is the connection to Collatz, the fact that the smallest version of the method suffices, and the machine-checked proof.

6 Rung two, begun: the “don’t repeat yourself” rule

There is a second result in the paper, of a completely different flavour, and it needs no power series at all.

The prefix-power rule. Suppose an orbit’s diary starts by repeating a block of ℓ letters over and over — say e times. Then the starting number and the number ℓ steps later have the same diary for $(e - 1)\ell$ steps, so they agree in their last $(e - 1)\ell$ binary digits. Two different whole numbers that agree in that many digits differ by at least $2^{(e-1)\ell}$; but ℓ steps of the game can only change a number by a factor of about $3^o/2^\ell$, where o counts the odd steps in the block. So

$$(e - 1)\ell \lesssim o \log_2 3 + \log_2 n.$$

A diary cannot start with a long high power of a short block unless the starting number is enormous. This is a three-line theorem in Lean.

Now, the threshold: if the block has ones-density δ , the rule bites when $e > 1 + \delta \log_2 3$. At the 63% line that is exactly $e = 2$. Above the line one needs a bit more than a square.

Enter the *Sturmian* diaries: the most orderly non-repeating words there are, the ones you get by writing 1 each time a point moving around a circle by a fixed irrational angle crosses a mark. They have a famous property: they almost repeat with period q_k for every denominator q_k of the continued fraction of the angle, and the almost-repetition lasts for about $q_k + q_{k+1}$ letters — a power with exponent 2.6 or more, comfortably above the threshold. Feeding that into the rule, with careful bookkeeping of how the orbit grows before the repetition starts, gives the paper’s second theorem: no orbit has a Sturmian diary whose angle has partial quotients ≥ 2 infinitely often (for angles up to 0.789; larger partial quotients handle larger angles). The bookkeeping is machine-checked; the two facts about circle rotations it relies on are classical (Lagrange, 1770s; Sós, 1958) and are stated as hypotheses rather than proved in Lean. Numerically the rule works for the remaining “golden” angles too, up to about 0.9.

It keeps happening that the 63% line shows up as the exact threshold of every mechanism we try. That is probably not a coincidence.